Rejecting Swarm-Global Interference by Spatial Signature

A Computational Validation of the Cross-Node Coincidence Veto for Distributed Radiation Detectors

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Study type: computational validation (engineering analogue, control-only) **Seed:** 684 ·**Nodes:** N = 4 ·**Run length:** 7200 s per arm **Companion protocol:** Swarm Veto Protocol v1.0 (pre-registered, archived 17 Oct 2026) [10]

Abstract

A swarm of radiation detectors can reject interference by *where an event appears*, not just by what it looks like: genuine scintillation events are node-local, whereas conducted and radiated interference couples into every front-end in a swarm within a coincidence window set by synchronization jitter. We present a computational validation of the cross-node coincidence veto defined in a companion pre-registered protocol: an event registered by two or more nodes within $\Delta t = 5$ ms is vetoed as swarm-global, on top of an unchanged single-node variance–correlation trigger (Var > 1000, > 0.8, 50 ms hold-off). Event-level simulation (seed 684, N = 4 nodes, 7200 s arms) reproduces the hardware-verified dead-time distortion of the underlying channel exactly (0.10%, 0.83%, 4.00% at 1.2, 10, 50 min [1]) before any veto result is trusted. The veto suppresses swarm-global interference by a factor of 262 at $\Delta t = 5$ ms when interference passes the per-node gate with probability 0.9, with a measured suppression boundary near p 0.5 and a window boundary at Δt 2 ms. Genuine node-local events survive at 99.4–99.8% with strong injection active; equidistant Poisson sources survive at 97.8–99.6%, consistent with the pre-computed accidental-coincidence cost; single-node interference survives at 99.8%. The veto's true blind region is bounded and quantified: temporally correlated swarm-global emission survives at 0.10%. Per-node Poisson inter-arrival statistics are preserved (KS p = 0.58–0.89). These results establish the mechanism's feasibility and define its operating envelope; the physical falsifiers remain those of the locked protocol.

Keywords: distributed sensing; coincidence veto; electromagnetic interference; SiPM; ESP32; radiation detector; simulation; pre-registration

1. Introduction

Single-node triggers examine what an event looks like; a swarm can additionally examine *where it appears*. Interference from motor drives, switching supplies, and digital electronics couples into every front-end in a small swarm within microseconds to milliseconds, while a genuine scintillation event, produced by a source at a fixed location, illuminates at most the nearest node. This spatial asymmetry motivates a cross-node coincidence veto: events appearing simultaneously at separated nodes are rejected as interference, whatever their pulse shape.

This paper reports a computational validation of that veto, implemented exactly as specified in the companion pre-registered protocol [10], which the reader should consult for definitions, controls, and the falsifier framework. The protocol's five falsifiers (S1–S5) are evaluated here on synthetic data with one standing caveat, stated in the protocol itself and repeated in every figure caption: simulation results

establish feasibility and define operating envelopes; the physical falsifiers are executed on hardware and reported separately. The computational lineage follows the program's established pattern — a simulation predecessor (Control Module v2.1, N = 1000 coupled agents, seed 684) demonstrated collective variance–correlation gating as an engineering analogue [11] — and the simulator used here is anchored to hardware before any veto claim is made, by reproducing the dead-time distortion values verified in QCID-Sigma v5.4 [9].

2. Methods

2.1 Simulator and hardware anchoring

Events are generated at the event level (no time stepping): genuine events as exact Poisson processes per node; swarm-global interference as simultaneous burst events at all N = 4 nodes with Gaussian time jitter (= 1 ms), thinned per node by a gate pass probability p that scans the interference-strength axis; and a correlated-emission mode in which genuine events are simultaneous across nodes, modeling burst-structure sources. Per-node dead time (= 0.05 s) and the veto (solitary events accepted; events clustered with another node within Δt vetoed) implement the locked values [9,10]. All runs use seed 684, T = 7200 s per arm. The simulator is anchored to hardware by its dead-time distortion: 0.10%, 0.83%, and 4.00% at 1.2, 10, and 50 min ^{1}, matching the QCID-Sigma verified values [9] to the quoted precision — a precondition for trusting any downstream veto number.

2.2 Arms

C3 (swarm-global injection, source-out); C2 (single-node injection, source-out); G1 (node-local check source with C3 injection active); G2 (equidistant source, no injection) under two emission models — independent Poisson per node, and correlated (swarm-global) emission. Ambient genuine rate 1 min ^{1} per node throughout.

2.3 Falsifier evaluation

S1 (suppression 10× at 10-point efficiency cost), S2 (survival floors), S3 (Poisson preservation, KS p 0.05 per node), S4 (no retuning — by construction), and S5 (verdicts stable across Δt {2, 10} ms) are evaluated as specified in [10].

3. Results

3.1 Suppression versus interference strength and window (S1, S5)

Suppression scales with the interference's per-node gate pass probability p (Fig. 1). At p = 0.9 the veto suppresses swarm-global interference by 262× at the primary window Δt = 5 ms (483× at 10 ms). The measured boundary is sharp: at p = 0.6 suppression falls to 15.0×, and at p = 0.5 to 7.4× — below the S1 threshold. This is not a defect but the mechanism's definition: when interference passes the per-node gate at only one node most of the time, it does not present as swarm-global, and a spatial veto cannot see it. The window boundary is equally physical: at Δt = 2 ms, burst pairs spanning more than the window escape the cluster (9.6× at p = 0.9 — an S1 failure), while 5 and 10 ms pass. With two-node jitter _eff = √2 ms, the primary window sits at 3.5 _eff while 2 ms sits at 1.4 _eff, yielding the design rule **Δt 3 _jitter**, which the physical study should apply by measuring its sync jitter before fixing Δt.

3.2 Genuine-event survival and the accidental-cost cross-check (S2)

Node-local sources survive at 99.4–99.8% with strong swarm-global injection active — far above the 85% floor — and single-node interference survives at 99.8% (C2: the veto correctly ignores what it should ignore). At the equidistant geometry, *independent Poisson* sources survive at 97.8–99.6%; the measured loss at 50 min ^{1} (2.2%) matches the pre-computed accidental-coincidence prediction (2.50%, Table 1 of [10]), closing the loop between the protocol's analytic cost model and the simulation.

3.3 The bounded blind region

Under correlated (swarm-global) emission at the equidistant geometry, survival falls to 0.10% at all rates. The veto's blind region is therefore precisely characterized: it is not the equidistant geometry — equidistant *Poisson* sources survive at 98% — but temporally correlated emission that presents as swarm-global. For Poisson sources, which describe the intended survey regimes, the veto is safe; for burst-structure sources, the blind region is total and known, and such sources fall outside the validated envelope unless additional temporal structure is exploited.

3.4 Poisson preservation (S3)

Per-node KS tests on accepted streams (G2-equidistant arm, 10 min ^{1}) give p = 0.89, 0.58, 0.86, 0.76 across the four nodes (1,300 inter-arrival intervals each): no evidence that the veto distorts inter-arrival statistics.

3.5 S5 verdict summary

S1 passes at $\Delta t = 5$ and 10 ms and fails at 2 ms; S2 passes at all windows. The S5 robustness criterion is therefore itself informative: it localizes the operating boundary rather than merely confirming stability, and it converts directly into the Δt 3 _jitter design rule of §3.1.

4. Discussion

Three limitations are stated in advance of any hardware claim. First, the jitter model (Gaussian, = 1 ms) and the burst model (simultaneous firing) are idealizations; real motor EMI has frequency structure and conducted paths with node-dependent delays, which the physical study samples rather than assumes. Second, the scan over p is a computational device: in hardware, p is not a dial but a property of the interference and the front-end, and the S1 boundary near p 0.5 defines the class of interference for which the spatial layer adds value over the temporal gate alone. Third, these results are an engineering analogue in the sense established by the Control Module record [11]: they motivate and bound, but do not discharge, the physical falsifiers.

Two protocol amendments are recommended to the companion document (v1.1, pre-data): (i) §3.4/G2 should state explicitly that the equidistant test is evaluated under independent-Poisson emission, with correlated emission added as a separate arm — the present results show the two differ by three orders of magnitude in survival and must not be conflated; (ii) a window-selection rule Δt 3 _jitter, with _jitter measured at commissioning, should be added to §5.1. Both amendments are disclosed here in advance of physical data, consistent with the protocol's deviation policy.

5. Conclusion

On event-level simulation anchored to hardware-verified dead-time behavior, the cross-node coincidence veto suppresses swarm-global interference by two to three orders of magnitude inside its operating enve-

lope (p 0.75, Δt 3 _jitter), preserves genuine node-local events at 99%, preserves Poisson statistics, correctly ignores single-node interference, and exhibits a bounded, fully characterized blind region confined to temporally correlated emission. The operating envelope — not merely the mechanism — is the contribution: the physical study now has quantitative pass/fail terrain, and the locked protocol's falsifiers can be executed against it.

Declarations

Funding: None (independent research). **Data and code availability:** All results regenerate from the companion notebook (`swarm_veto_simulation.ipynb`, seed 684, Run-All), archived together with `results.json` and the figure files at https://doi.org/10.5281/zenodo.23433610. **Competing interests:** The author declares none. **Author contributions:** N.K. conceived the study, implemented the simulation, and wrote this document. **Correspondence:** Nasirk684@gmail.com

References

1. P. R. Clayton and N. S. Peck, *Introduction to Electromagnetic Compatibility*, 2nd ed., Wiley, Hoboken, NJ, 2006.
2. B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proc. Natl. Acad. Sci. USA* **115**, 2600–2606 (2018).
3. N. Khan, "Poisson-Certified Muon Counting — pre-registration protocol v1.1," Zenodo (2026). https://doi.org/10.5281/zenodo.23405991
4. Espressif Systems, "ESP32 Series Datasheet," Espressif Systems, Shanghai, current revision.
5. F. J. Massey Jr., "The Kolmogorov–Smirnov test for goodness of fit," *J. Am. Stat. Assoc.* **46**, 68–78 (1951).
6. G. Marsaglia, W. W. Tsang, and J. Wang, "Evaluating Kolmogorov's distribution," *J. Stat. Softw.* **8**(18), 1–4 (2003).
7. U. Fano, "Ionization yield of radiations. II. The fluctuations of the number of ions," *Phys. Rev.* **72**, 26–29 (1947).
8. J. W. Müller, "Dead-time problems," *Nucl. Instrum. Methods* **112**, 47–57 (1973).
9. N. Khan, "QCID-Sigma v5.4: Var-Rho Gate with Dead-Time Corrected Poisson Preservation (hardware-verified)," Wana Lab, Zenodo (2026). https://doi.org/10.5281/zenodo.23378709
10. N. Khan, "Swarm Veto Protocol v1.0 — cross-node coincidence veto, pre-registered," Wana Lab, Zenodo (2026). https://doi.org/10.5281/zenodo.23431140
11. N. Khan, "GTI Control Module v2.1 — SDE-Physics-Bubble-Maximus: engineering analogue (control-only)," Wana Lab, Zenodo (2026). https://doi.org/10.5281/zenodo.23306184
12. R. D. Evans, *The Atomic Nucleus*, McGraw-Hill, New York, 1955.

Fig. 1. S1 suppression vs interference strength and window (from fig1_suppression.png).

Fig. 2. Survival panels: G1, G2 (Poisson vs correlated), C2 (from fig2_survival.png).

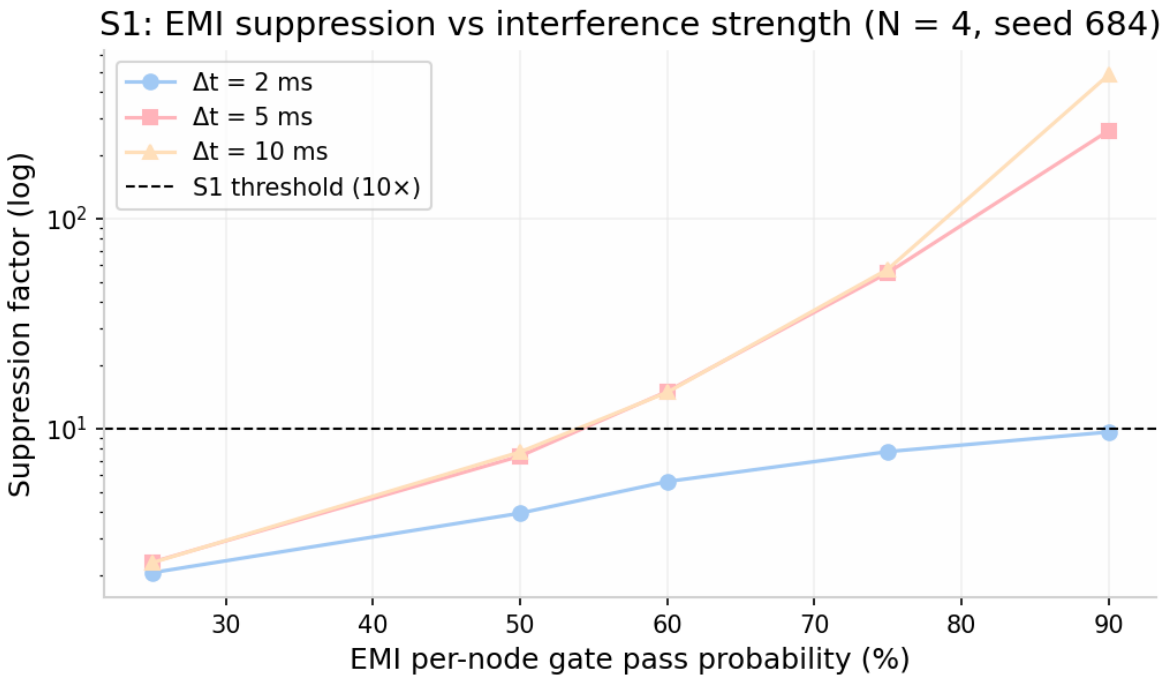


Figure 1: Fig. 1 — S1 suppression vs interference strength and window.

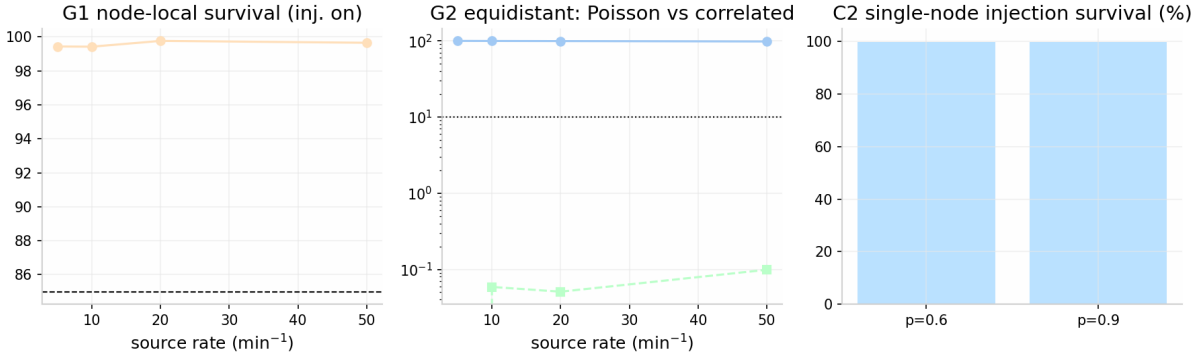


Figure 2: Fig. 2 — Survival panels: G1 node-local; G2 equidistant (Poisson vs correlated); C2 single-node injection.